BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering
Level-3, Term-I Final Examination:June 2016

Subject: CSE 305 (Microprocessors and Microcontrollers)

Time: 3.00 hours **Full Marks: 210**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. Answer any **THREE** questions from each section.
c. Figures in the margin indicate full **marks**.
d. Assume reasonable data if necessary.
e. **Symbols** used have their usual meanings.

SECTION-A

Question – 1

a. What are the basic differences between microprocessor and microcontroller? **06**

b. Write down an assembly program for printer driver procedure (mode 1), which is connected with an 8086 microprocessor via 82C55 (PPI). Here, the printer is connected with 82C55 through port-A. 82C55 PPI is decoded at I/O port number 0BH to 0EH. [See the necessary registers at the end, Figure:1] **12**

c. What is the main purpose of Direct memory access (DMA)? Describe different types of direct memory access. **2+9=11**

d. Draw the internal structure of 8254. **06**

Question – 2

a. Describe the working principle of line decoding method and also state the disadvantages of this method. **5+5=10**

b. Draw an 8-bit microprocessor interfaced with a 5K RAM system using fully decoded addressing technique. The microprocessor has 10 bit address bus. **08**

c. Draw a flowchart of keyboard scanning procedure. **05**

d. Write down the differences between the followings : **4×3=12**
   i. Full and partial decoding method.
   ii. 2-Key lockout and N-Key rollover method.
   iii. Mode 0 and mode 1 operation of 8254.

Question – 3

a. What is the main purpose of control unit in a microprocessor? Discuss the techniques of designing the control unit. **09**

b. Write down a program for 8086 microprocessor to write a 8-digit numerical display with right entry which is interfaced with an 8279 Programmable Keyboard/Display. Interface (PK/DI) at I/O port address 80H (for data) and 81H (for control/status). The 8279 CLK pin is connected to a 3.0 MHz clock. Assume, the keyboard uses encoded N-key rollover method. [See the necessary registers at the end, Figure: 2]. **12**

c. Define standard I/O and memory mapped I/O with suitable example. **08**

d. Discuss the command byte of 82C55 PPI. **06**

Question – 4

a. What is interrupt driven I/O and how it works? **08**

b. Suppose, a programmable interval timer 8254 is connected to function at I/O ports 0700H, 0702H, 0704H, 0706H of a 8086 microprocessor. Counter 0 operates in Mode 3 (square wave) and counter 2 operates in Mode 2 (continuous wave). The count programmed in counter 0 is 60 and the count of counter 2 is 305. All counts are written in binary format. Write an initialization program for 8254. [See the necessary registers at the end, Figure: 3] **10**

c. Draw the internal structure (both Port-A and Port-B) and timing diagram of strobed input operation of the 82C55. **10**

d. What are the advantages of 8279 PK/DI over 82C55 PPI. **05**

SECTION-B

Question 5

a. Develop a procedure for arithmetic coprocessor 80×87 that will calculate the area of a triangle and the result must be saved in a variable called triangle_area. Also show the status of the stack of 80×87 after each instruction is executed. [Area of a triangle $=\frac{1}{2}\times$ height $\times$ base, suppose here height = 2.5m and base = 12.0m. **10+5=15**

b. Differentiate between the direct and indirect modes for accessing I/O ports with example. **04**

c. What main function is provided by 8288 bus controller when used with 8086/8088? **06**

d. Compare among 8086, 80186, 80286 microprocessors. **10**

Question 6

a. Discuss what will happen if a higher priority interrupt comes to an unmasked interrupt request pin of 8259A when 8086 is already executing a lower priority. **12**

b. Suppose, 8086 microprocessor executes the instructions: **5+5=10**

i. MOV ARRAY[SI] , CX ; Here [SI] = 000AH and [DS] = EEEEH and ARRAY = E2H. Now calculate the 20-bit memory address where the processor writes the data of CX register.

ii. MOV BYTE. Here, [DF] = 1, [DS] = 2000H, [SI] = 0500H, [ES] = 4000H, [DI] = 0300H. Now calculate memory address where the processor moves data.

c. Draw the connection diagram of 16550 with microprocessor. **08**

d. How does poll and poll status register of 80186 differ from each **05**

other?

Question 7

a. What is the purpose of Tag Register in 80×87 coprocessor? Briefly discuss the six errors that are found in the 80×87 arithmetic coprocessor status register. [See the necessary registers at the end; Figure: 4(a-b)] **4+6=10**

b. Write down a program for 8086 microprocessor to program 8259A Programmable Interrupt Controller (PIC) and 16550 UART for receiving 6-bit serial data which sends 1200 bauds per second. Other bit pattern of baud frame is two stop bit, stick parity off, odd parity as parity and transmit no break between two bauds. The INTR signal from the 16550 UART is connected to the programmable interrupt controller's interrupt request pin IR3. 18.432 MHz crystal is used as a timing source in 16550 UART. The 8259A PIC is set up to function at interrupt vectors D0H-D7H and operates with automatic EOI. The 16550 UART is decoded at I/O ports 80H-87H, the 8259A PIC is decoded at I/O ports 31H and 35H. [See the necessary registers at the end; Figure : 5(a-c) and Figure : 6(a-c)] **20**

c. What is watchdog timer? **05**

Question 8

a. The address/data bus of 8086/8088 is multiplexed. Now discuss the demultiplexing process of address/data bus of 8086/8088. **15**

b. What do you understand by sending a break between two frames in UART? **05**

c. What is automatic END of interrupt? **05**

d. How does the instruction queue of BIU work in 8086. **10**

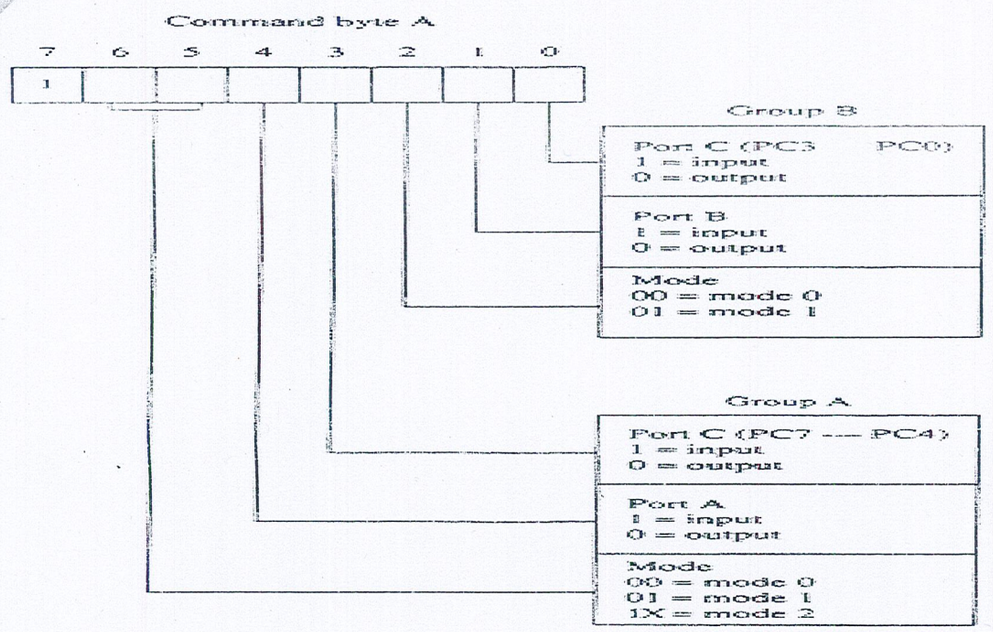

Figure-1 : Command Byte A of 82C55 (Figure for Question 1b)

DD	*Function*
00	8-digit display with left entry
01	16 digit display with left entry
10	8-digit display with right entry
11	16-digit display with right entry

MMM	*Function*
000	Encoded keyboard with 2-key lockout
001	Decoded keyboard with 2-key lockout
010	Encoded keyboard with N-key rollover
011	Decoded keyboard with N-key rollover
100	Encoded sensor matrix
101	Decoded sensor matrix
110	Strobed keyboard, encoded display scan
111	Strobed keyboard, decoded display scan

000DDMMM : Mode Select
001 PPPPP : Clock command
010 Z 0 AAA : Read FIFO
011 Z AAAA : Display read
100 Z AAAA : Write display
101 0WWBB : Display write inhibit
110 0 CCFA : Clear
111 E 000 : End of interrupt

Figure-2 : Registers and codes used for programming 8279(Figure for question 3b)

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

Figure-3 : Control Word Register of 8254(Figure for question 4b)

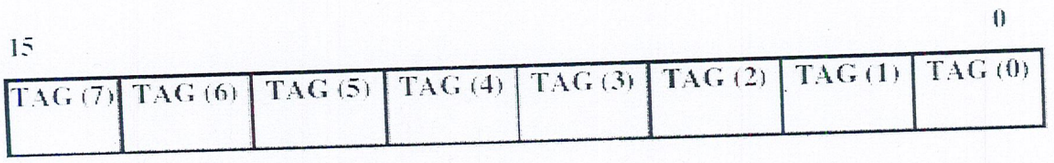

Figure-4(a): The Tag register of 80X87 (Figure for Question-7a)

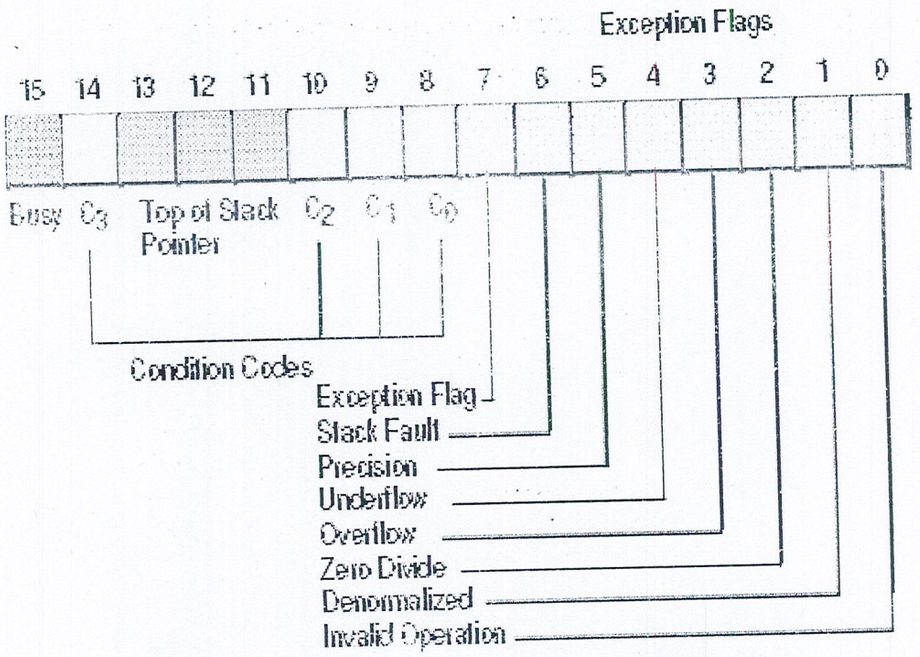

Figure-4(b): The status register of 80X87 (Figure for Question-7a)

A_2	A_1	A_0	*Register*
0	0	0	Receiver buffer (read) and transmitter holding (write)
0	0	1	Interrupt enable
0	1	0	Interrupt identification (read) and FIFO control (write)
0	1	1	Line control
1	0	0	Modem control
1	0	1	Line status
1	1	0	Modem status
1	1	1	Scratch

Figure-5(a): The registers of 16550 UART selected by A0, A1 and A2(Figure for Question-7b)

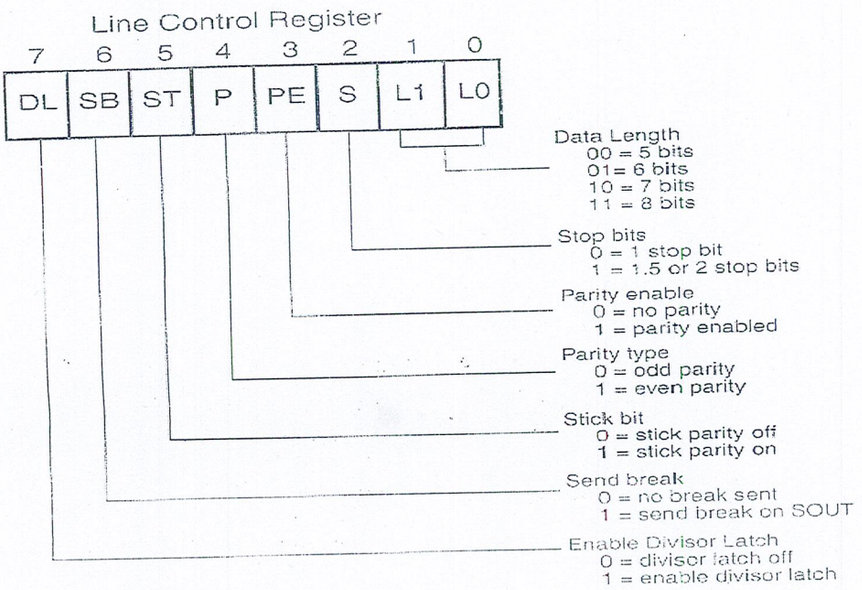

Figure-5(b): Line Control Register of 16550 UART (Figure for Question-7b)

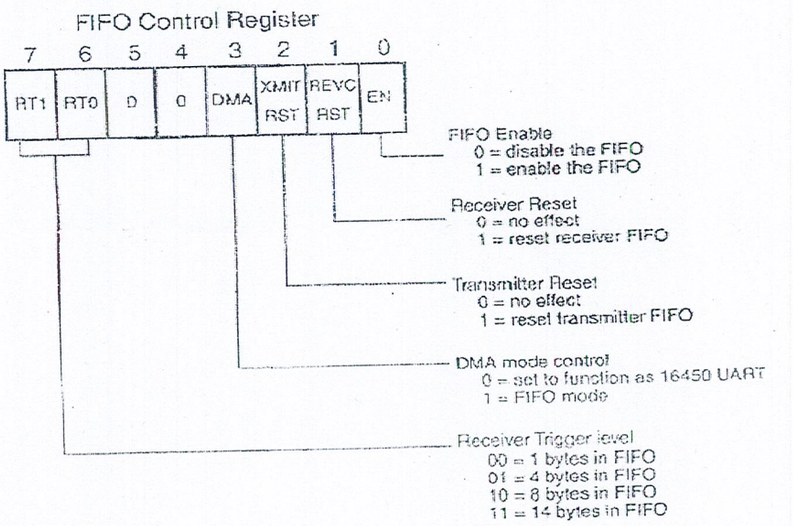

Figure-5(c): FIFO Control Register of 16550 UART (Figure for Question-7b)

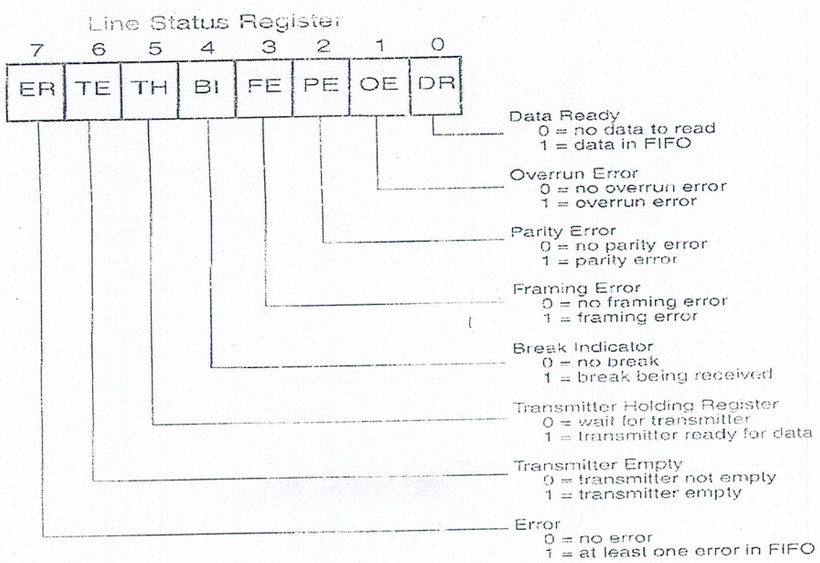

Figure-5(d): Line Status Register of 16550 UART (Figure for Question-7b)

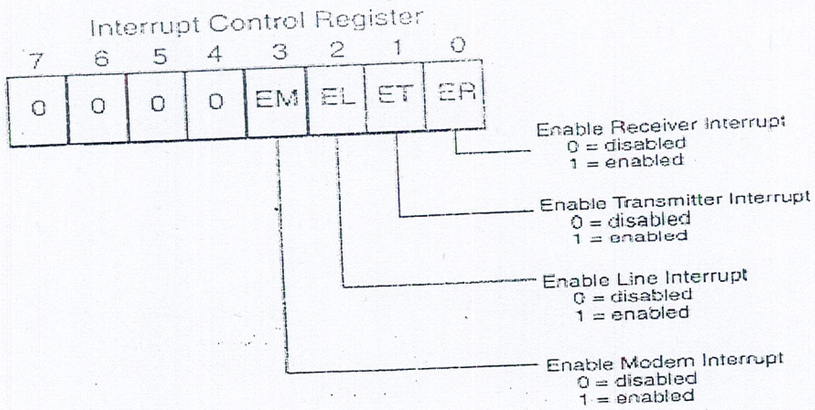

Figure-5(e): Interrupt Control Register of 16550 UART (Figure for Question-7b)

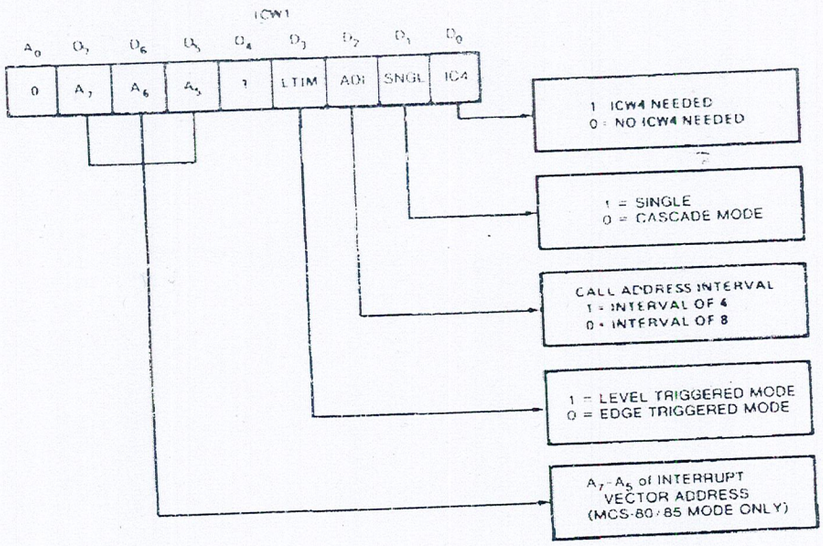

Figure-6(a): Initialization Command Word-1 (ICW1) of 8259A PIC (Figure for Question-7b)

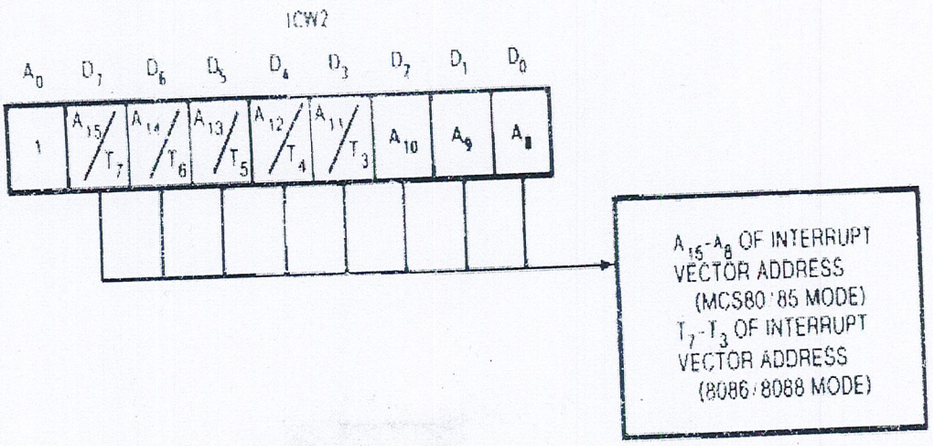

Figure-6(b): Initialization Command Word-2 (ICW2) of 8259A PIC (Figure for Question-7b)

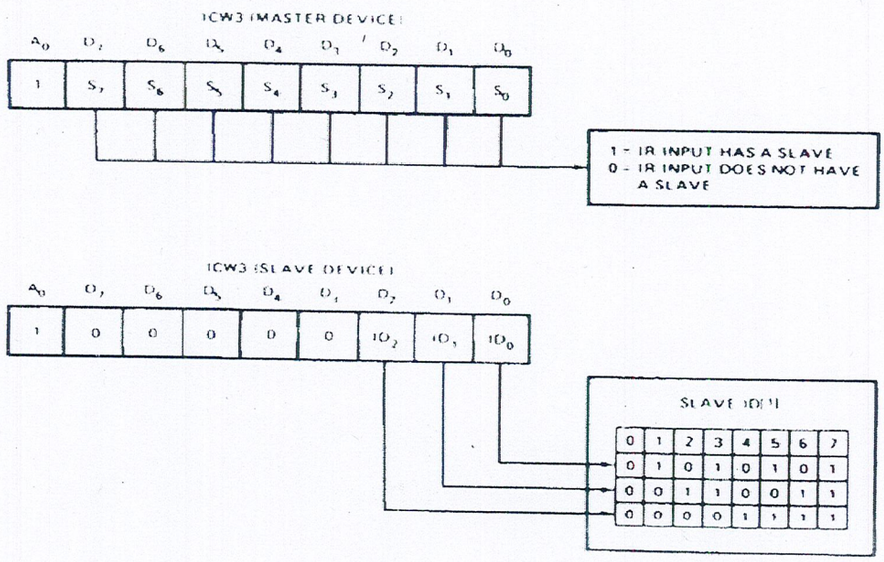

Figure-6(c): Initialization Command Word-3 (ICW3) of 8259A PIC (Figure for Question-7b)

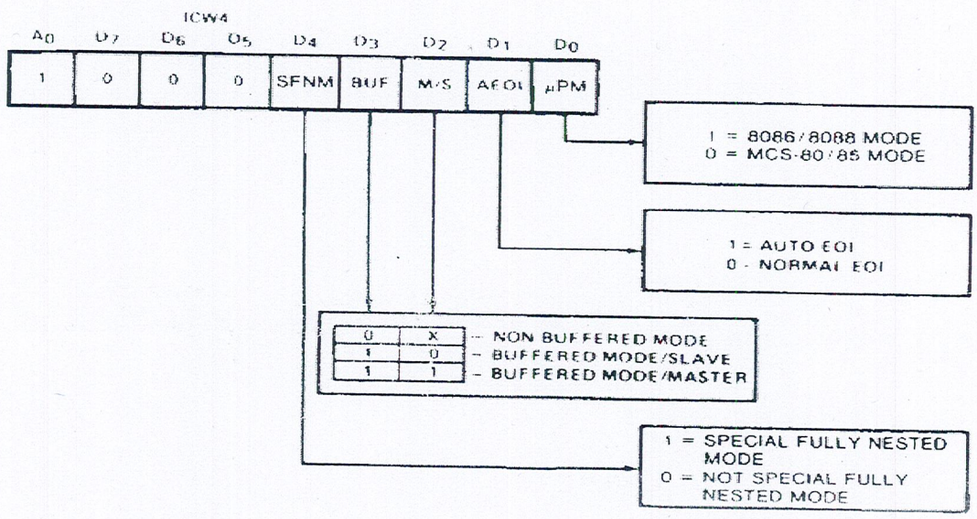

Figure-6(d): Initialization Command Word-4 (ICW4) of 8259A PIC (Figure for Question-7b)

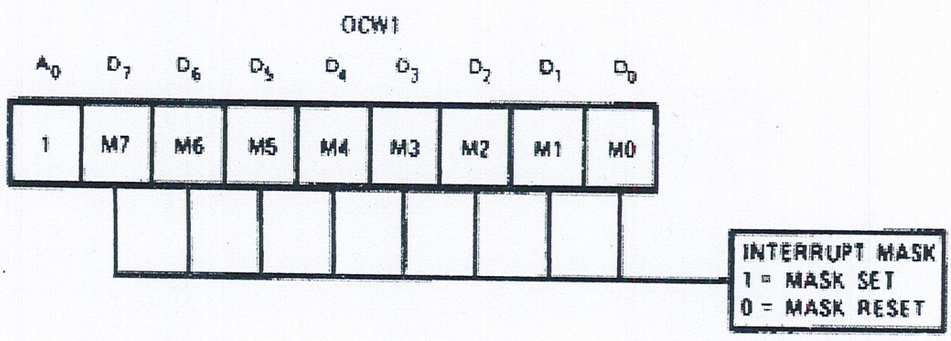

Figure-6(e): Operation Command Word-1 (OCW1) of 8259A PIC (Figure for Question-7b)

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INSTRUCTIONS:

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SECTION-A

Question – 1

a. Write down the basic differences between microprocessor and microcontroller. **5**

b. Consider the following Figure- 1: **12+6=18**

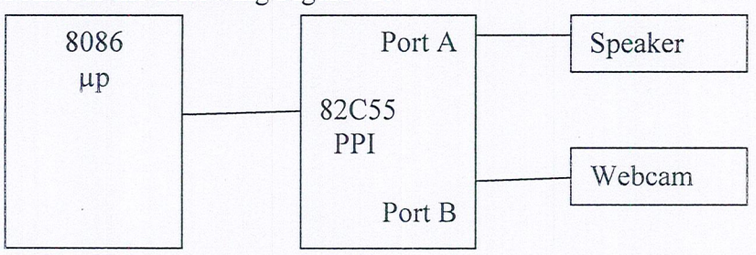


Figure 1: Peripheral devices connected with 8086 Microprocessor through 82C55 PPI.

Suppose a speaker and a webcam are connected with an 8086 microprocessor via 82C55 PPI through Port A and Port B respectively. 82C55 PPI is decoded at I/O port number 0BH to 0EH.
[See the necessary registers at the end: Figure-3]

i) Write an assembly program for Webcam and Speaker driver procedure where the webcam and speaker both work in strobe mode.
ii) Draw the timing diagram of Webcam driver procedure mentioning the signals needed to complete the process.

c. What are the advantages of 8279 PK/DI over 82C55 PPI. **4**

d. Draw the internal diagram of 8254. **8**

Question – 2

a. Discuss the multifunctional port line of 8051 microcontroller. **5**

b. Suppose, you have a microprocessor of 16 bit address bus and data bus of 8 bit and some RAM of each 256B. **8+8+6=22**

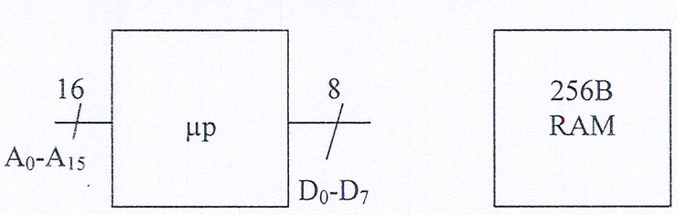


Figure 2: Microprocessor and RAM chip

Draw a neat diagram showing the connections between the above microprocessor and RAM to include the memory map 0400_{16} through $04EE_{16}$

i) Use line decoding method.
ii) Use Partial decoding method.
iii) What is the maximum number of RAMs you can associate with this microprocessor in both the methods?

c. Why "Wait State" is provided in 8279 PK/DI? Suppose, the 8279 CLK pin is connected to a 2.5 MHz clock. What will be the value of prescaler? **3+5=8**

Question – 3

a. What is bit addressability and how do this feature help 8051 microcontroller to perform better than those which does not support this feature? **5**

b. Suppose, a programmable interval timer 8254 is connected to function at I/O ports 0108H to 010BH of a 8086 microprocessor. Counter 0 operates in mode 3 (Square Wave) and counter 2 operates in mode 2 (Continuous Wave). The count programmed into counter 0 is 75D and the count of counter 2 is 562D. Write an initialization program for 8254. [See the necessary registers at the end Figure-4.] **12**

c. Why Programmable Peripheral Interface is needed to associate with microprocessor? Discuss the command bytes of 82C55 PPI. **2+8=10**

d. Suppose, you and your friends are going to a theme park and about to enjoy one of our favorite rides "Boat Splash" which allow only 5 persons to ride on a single boat. The gate of the ride will be opened only when the ride- controller pushes "OPEN" button. Then it will start to take 5 persons and when the last member enters, the gate is automatically closed so that no one can intrude into the ride.
If this system is implemented using 8254, in which mode it should be designed to achieve the above functionality. Also write down the basic features of that mode. **8**

Question – 4

a. Discuss the RAM organization of 8051 microcontroller. Also show the decomposition of internal RAM with associate addresses. **12**

b. Design a 96-Key keyboard interface to 8088 microprocessor through 8279 PK/DI. **9**

c. What is the main purpose of Direct Memory Address? Describe different types of direct memory access. **1+9=10**

d. How many ways the Keyboard can be interfaced with the processor. Briefly discuss the methods. **4**

SECTION-B

Question 5

a. Suppose, three interrupt request pins (IR2, IR5 and IR7) of 8259A, programmable interrupt Controller, are unmasked. An universal asynchronous receiver transmitter, 16550 is connected to IR2. Now, a divide error interrupt occurs at IR7 and 8259A sends the interrupt to 8086. Discuss what will happen if 16550 gives an interrupt, when 8086 is already executing the previous interrupt. **12**

b. What do you understand by high impedance state of address/data bus during hold acknowledgement of 8086 microprocessor? **8**

c. Write a procedure in assembly language for 16550, universal asynchronous receiver transmitter, that will receive data and check for errors. If there is any parity error, CL register will contain ASCII of 'P', if there is overrun error while receiving data, CL register will contain ASCII of 'O', otherwise, it will transfer received data to AL register. [See the necessary registers at the end-Figure: 5(a-b)]. **10**

d. What is the difference between non-specific end of interrupt and rotate on non-specific end of interrupt? **5**

Question 6

a. How do bus unit, instruction unit, execution unit and address unit of 80286 work together? **8**

b. Suppose, two programmable interrupt controller, $8259A_s$ are connected with each other where the slave device is connected to interrupt request 5 of master device. Now draw the connection diagram between these two devices. **7**

c. Discuss about the stack addressing mode and register addressing mode of 80 x 87 with example. **4+4=8**

d. Suppose, 8086 microprocessor executes the following instructions. Find the effective address and describe what happens after executing the instructions. **6+6=12**

i) Mov ARRAY [BP] [DI], AH; Here [DI] = $000A_{16}$, [BP] = 1230_{16}, ARRAY = $E2_{16}$, [DS] = 2000_{16} and [SS] = 2040_{16}.

ii) STOSB; Here, [SI] = 0002_{16}, [DI] = 2003_{16}, [ES] = 0300_{16}, [AL] = 08_{16}, DF = 1 and the address pointed by E.S : DI contains 42_{16}.

Question 7

a. How does poll and poll status register of 80186 microprocessor differ from each other? **5**

b. Suppose, a microprocessor has 24 bit address bus (A0- A23) and 16 bit data bus (D0-D15). The lower 12 bit of data bus is multiplexed with lower 12 bit of address bus (AD0-AD11) and the upper 4 bit of data bus (D12-D15) and upper 12 bit of address bus (A12-A23) work individually. Now design a fully buffered system for this microprocessor. **10**

c. Develop a procedure for arithmetic coprocessor 80 x 87 that will calculate the solution of the equation – **10+5=15**

$$(a+b)^2 = a^2 + b^2 + 2ab.$$

Also show the status of the stack of 80 x 87 after each instruction is executed.

[Suppose, a = 234.56 and b = 588.2345]

d. Draw the connection diagram of 16550, universal asynchronous receiver transmitter with microprocessor. **5**

Question 8

a. Write down a program for 8086 microprocessor to program 8259A, programmable interrupt controller and 16550, universal asynchronous receiver transmitter. **20**

i) The 16550 receives 8- bit serial data which sends 300D bauds per second. Other bit pattern of baud frame is two stop bits, no parity bit and a break between two data frames.

ii) The INTR signal from 16550 UART is connected to the 8259A's interrupt request input IR6.

iii) 18.432 MHz crystal is used as timing source in 16550 UART.

iv) The 16550 UART is decoded at I/O) ports 0E0H to 0E7H of 8086.

v) The 8259A PIC is set up to function at interrupt vectors 0A5H-0ACH and operates with rotate on non-specific end of interrupt, specially fully nested mode and will trigger on level triggered mode.

vi) The 8259A PIC is decoded at I/O ports 22H and 24H.

[See the necessary registers at the end: Figure- 5 (b-c) and 6 (a-f)].

b. Differentiate between direct and indirect modes for accessing I/O ports of 8086 microprocessor with example. **5**

c. Compare between 8086, 80186 and 80286 microprocessors. **10**

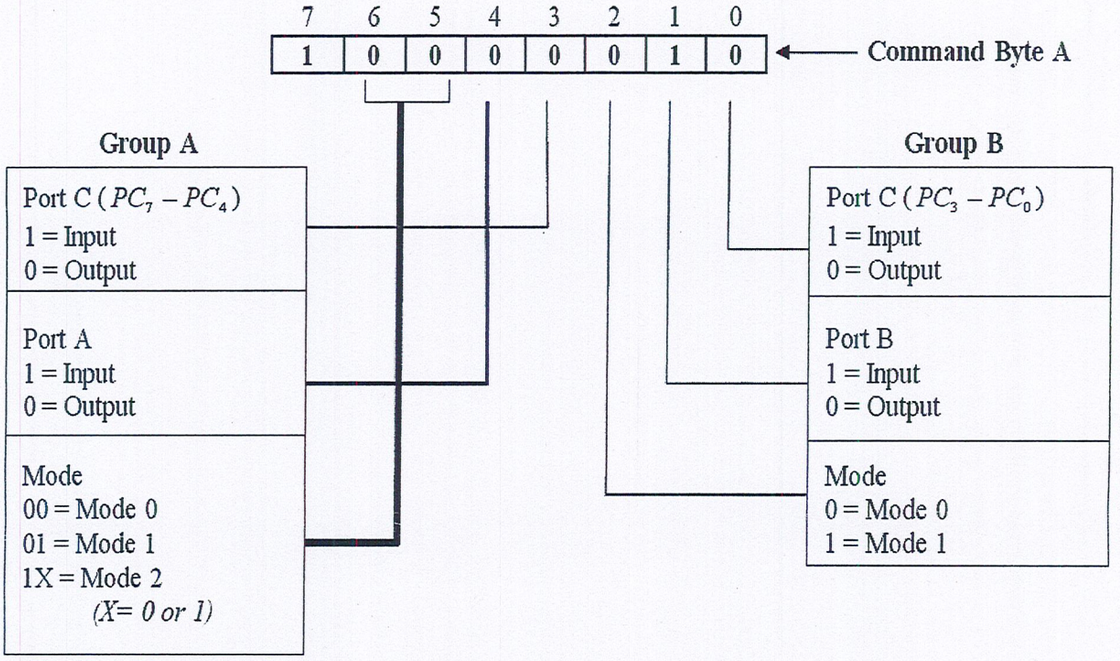


Figure-3 : Command Byte A of 82C55

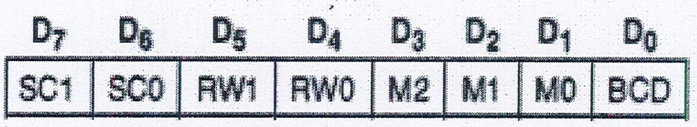


Figure-4 : Control Word Register of 8254

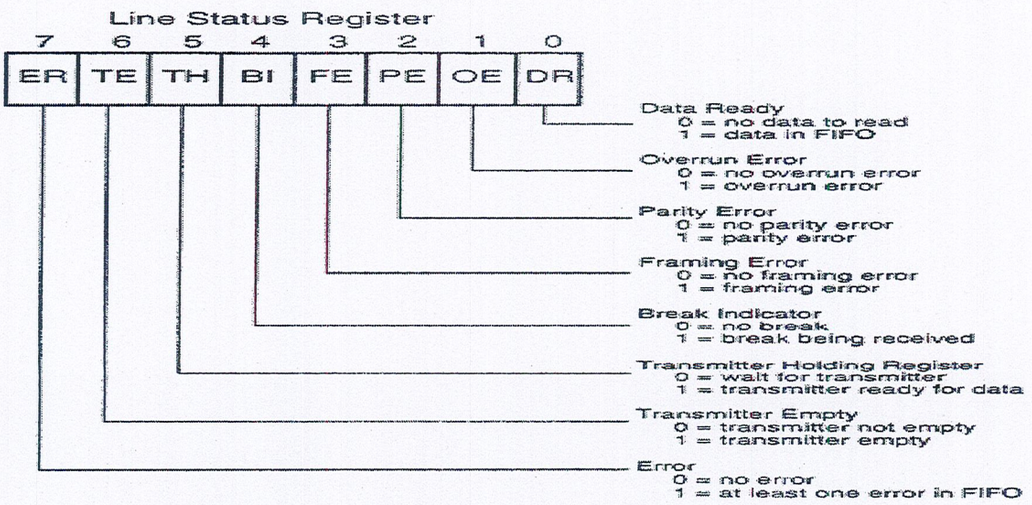


Figure-5(a): Line Status Register of 16550 UART (Figure for Question-5c)

A_2	A_1	A_0	*Register*
0	0	0	Receiver buffer (read) and transmitter holding (write)
0	0	1	Interrupt enable
0	1	0	Interrupt identification (read) and FIFO control (write)
0	1	1	Line control
1	0	0	Modem control
1	0	1	Line status
1	1	0	Modem status
1	1	1	Scratch

Figure-5(b): The registers of 16550 UART selected by A0, A1 and A2(Figure for Question-5c & 8a)

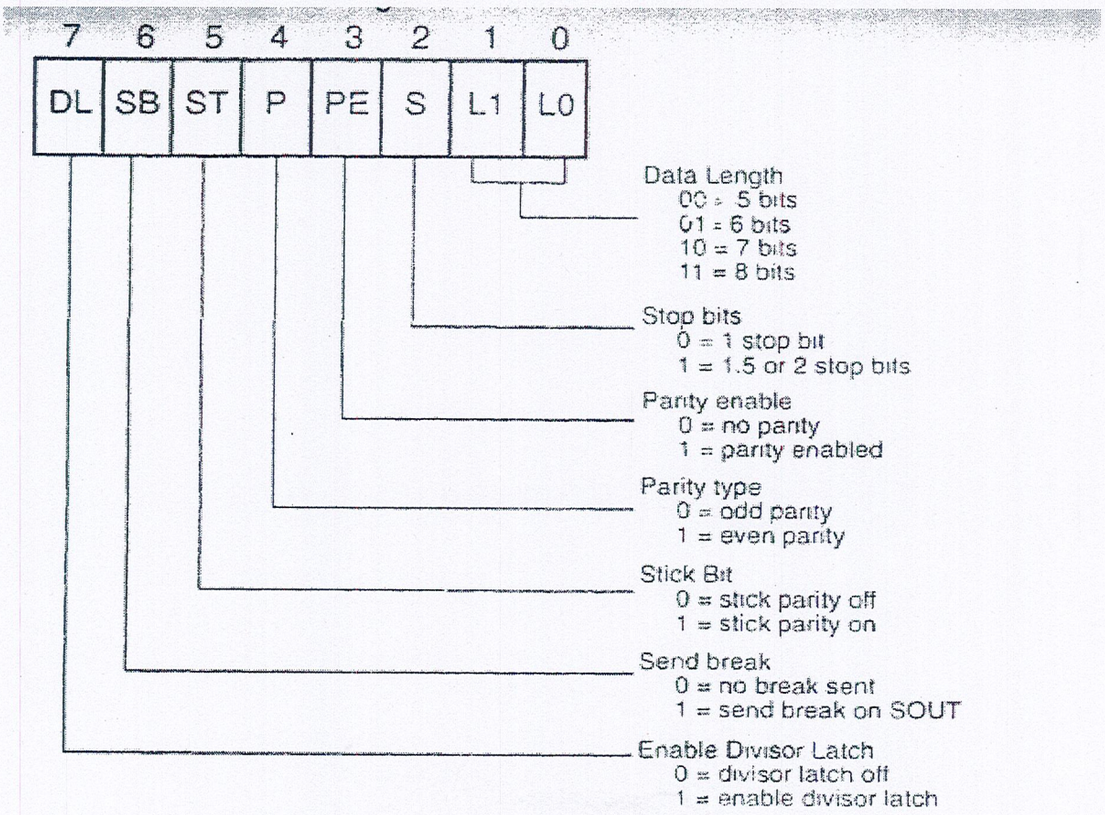


Figure-5(c): Line Control Register of 16550 UART (Figure for Question-8a)

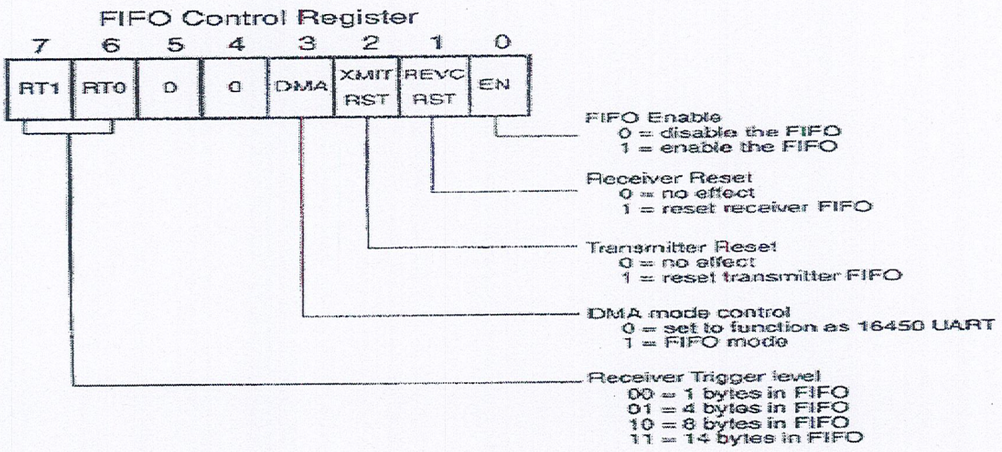


Figure-5(d): FIFO Control Register of 16550 UART (Figure for Question-8a)

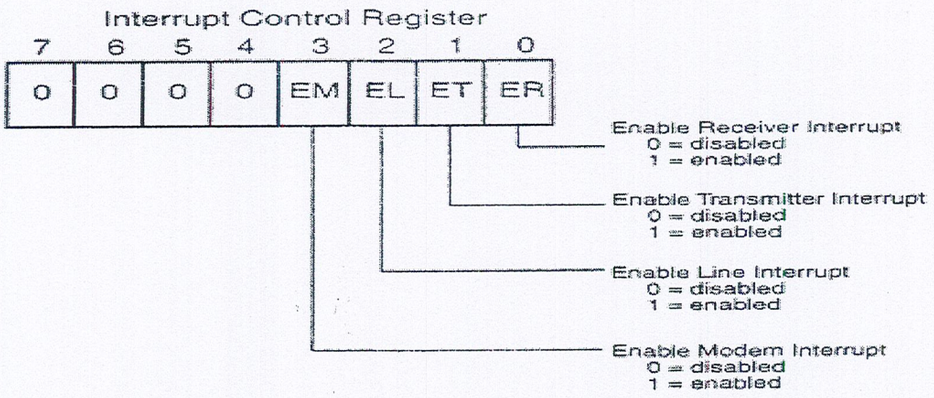


Figure-5(e): Interrupt Control Register of 16550 UART (Figure for Question-8a)

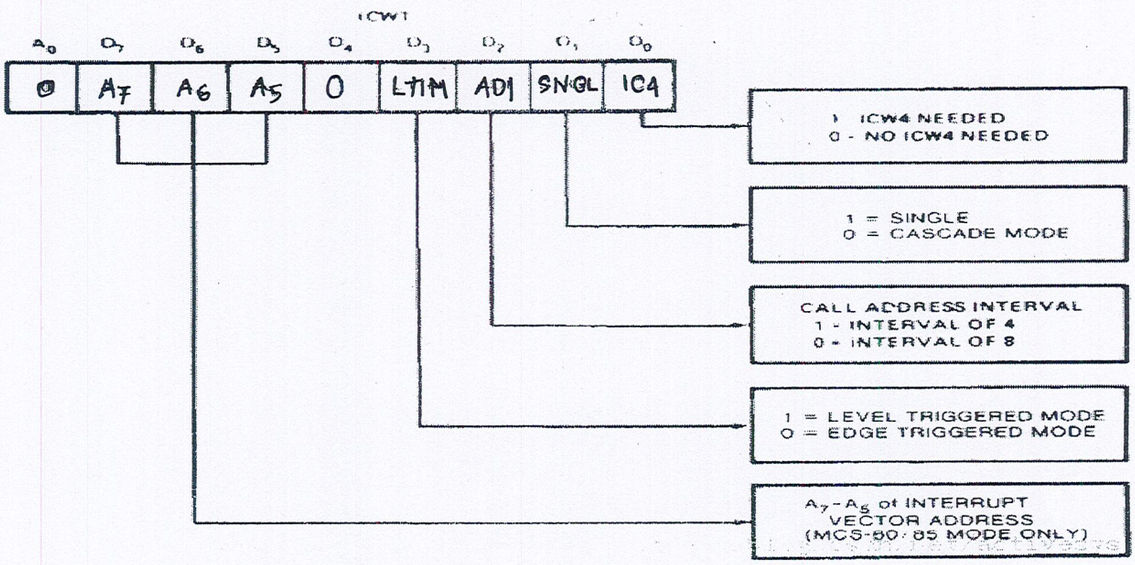


Figure-6(a): Initialization Command Word-1 (ICW1) of 8259A PIC (Figure for Question-8a)

Tabular

ICW_2

A_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	T_7	T_6	T_5	T_4	T_3	A_{10}	A_9	A_8

T_7-T_3 are $A_3 - A_0$ of Interrupt vector address
$A_{10} - A_9$, A_8 – Selected according to Interrupt request level.
They are not the address lines to microprocessor
$A_0 - 1$ Selects ICW_2

Figure-6(b): Initialization Command Word-2 (ICW2) of 8259A PIC (Figure for Question-8a)

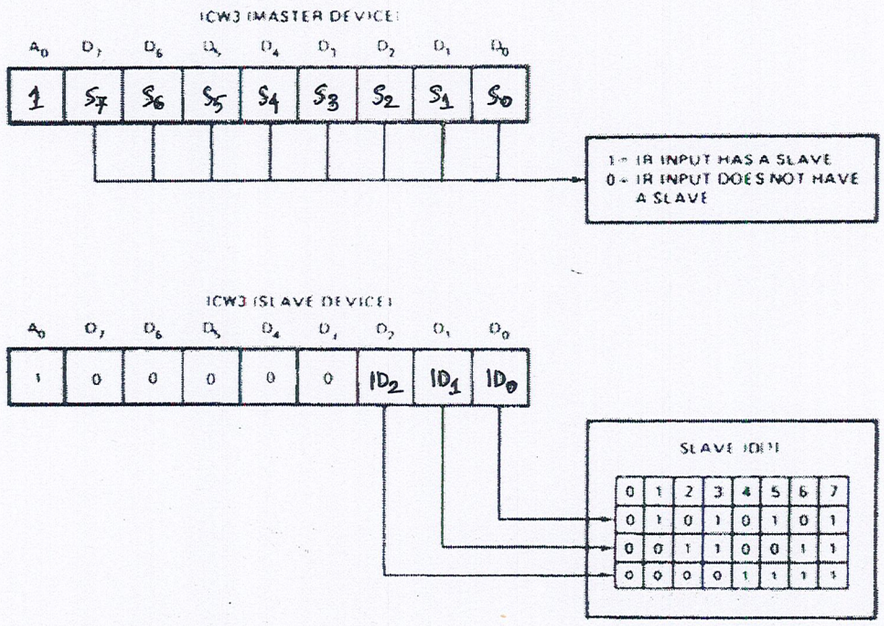


Figure-6(c): Initialization Command Word-3 (ICW3) of 8259A PIC (Figure for Question-8a)

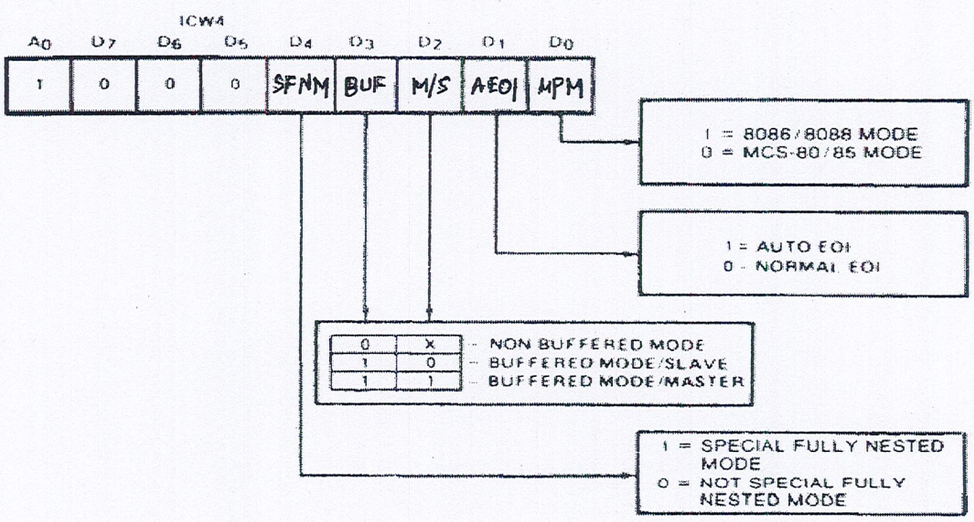


Figure-6(d): Initialization Command Word-4 (ICW4) of 8259A PIC (Figure for Question-8a)

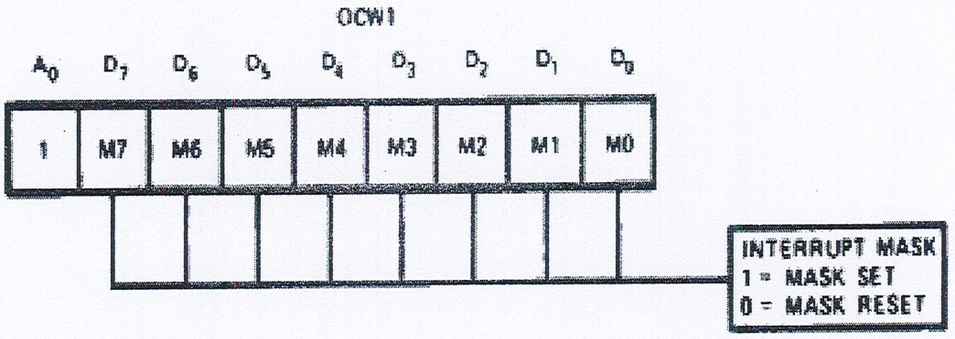


Figure-6(e): Operation Command Word-1 (OCW1) of 8259A PIC (Figure for Question-8a)

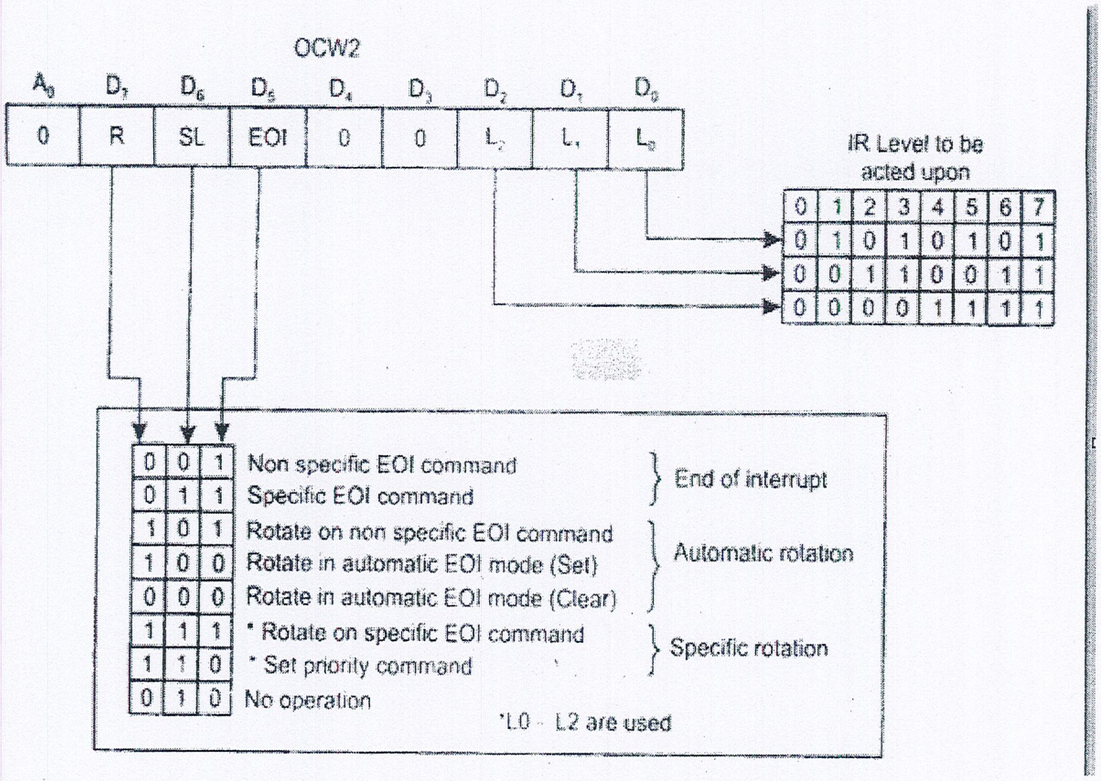


Figure-6(f): Operation Command Word-2 (OCW2) of 8259A PIC (Figure for Question-8a)

Tutorial

BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering
Level-3,Term-I Final Examination : June- July 2018

Subject: CSE-305, Microprocessor And Microcontrollers

Time: 3.00 hours **Full Marks: 210**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. Answer any **THREE** questions from each section out of **FOUR.**
c. Figures in the margin indicate full **marks**.
d. Assume reasonable data if necessary.
e. **Symbols** used have their usual meanings.

SECTION-A

Question – 1

a. Write down the basic differences between microprocessor and microcontrollers? Explain with suitable examples. **6**

b. Suppose, a touchscreen is connected with an 8086 microprocessor via 82C55 (PPI). The port A of 82C55 is configured as input port and Port B of 82C55 is configured as output port. 82C55 is decoded at I/O port number 2BH to 2EH. **14**
Write an assembly program for touchscreen driven procedure where the device works in strobe mode. [See the necessary diagram at the end: Figure-1]

c. "The 8086 can read a 16-bit word in one operation if the first byte of the word is at an even address. On the other hand, the 8086 must perform two memory accesses to two consecutive even addresses, if the first byte of the word is at an odd address" – Explain the above statement with suitable example. **9**

d. What is the difference between encoded and decoded keyboard as defined for 8279 Programmable keyboard/ Display Interface (PK/DI). **6**

Question – 2

a. Define register addressing mode and immediate addressing mode of 8086 micro-processor with suitable examples. What will be the value of AX register in each step after execution the following instructions? **4+5=9**

```
MOV AX, ALPHA [BX] [SI]
ADD AL, 03
INC AX
```

Where, [BX] = 0201H, ALPHA = 09H, [SI] = 1200H, DS = 300AH

b. What is the significance of FIFO status register of 8279 PK/ DI? Write down the assembly code that will find out the number of characters present in the FIFO on a specific time. [See the necessary diagram at the end : Figure-2] **5+10=15**

c. Suppose, you are going to design a Keyboard which will have the alphabets (English) and numeric characters. There will be a special key that will differentiate between the lowercase and uppercase characters. Now, interface the keyboard with 8088 microprocessor through 8279 PH/ DI. **6**

d. What is the importance of read back command in 8254 programmable Interval Timer? **5**

Question – 3

a. Draw the timing diagram of scanner driver procedure, mentioning the signals needed to complete the process. **5**

b. Suppose, timer/counter umber 1 (TM1) is configured in mode 0 and load it with 2 ms period at 1.5 MHz clock frequency. Time/counter number 2 (TM2) is configured in mode 2 and load it with 50 μs period at 3MHz clock frequency. 8254 is connected to function at ports 0101H to 0104H. All the counters are written in binary format.
Write an initialization program for 8254. [See the necessary diagram at the end: Figure:3] **14**

c. Why do we need to set command byte in 82C55 PPI? Briefly explain the command bytes of 82C55 PPI. **3+8=11**

d. Write down the functionality of the following signals of 82C55 PPI: **5**
   i. $\overline{STB}$
   ii. $\overline{ACK}$
   iii. IBF
   iv. $\overline{OBF}$
   v. $INTR$

Question – 4

a. Now-a-days, the terms CPU and Microprocessor have become synonymous. However, there still has a major difference between these two terms. Mention the differences between CPU and Microprocessor in terms of computer architecture. What does it mean when we say – "8086 is a 16 bit Microprocessor"? **3+2=5**

b. Write down the differences between mode 0 and mode 1 of 8254 with example. What is the significance of "GATE" signal in these two modes? **7**

c. Suppose, you have a microprocessor of 16 bit address bus and data bus of 8 bit and some RAMS, each of 512B. Draw a neat diagram showing the connections between the above microprocessor and RAMs to include the memory map 0800H through 08FFH. **6+6+6=18**
   i. Use line decoding method
   ii. Use partial decoding method
   iii. What is meant by fold back error and bus conflict error in Line decoding? How are these problems solved in partial/full decoding method?

d. Define different types of segmentation of 8086 microprocessor with suitable figure. **5**

SECTION-B

Question – 5

a. Describe the six modem control signals of 16550 UART. **6**

b. Describe the control & debug registers of 80386. **9**

c. Write down a program for 8086 microprocessor to program 8259A PIC and 16550 UART for 6 bit serial data which sends 1200 baud per second. Other bit pattern of baud frame is 1.5 stop bits, odd parity and a beak between two data frames. The INTR signal from 16550 UART is connected to 8259A's interrupt request input IR5. 73.728 MHz crystal is used as timing source in 16550 UART. The 16550 UART is decoded at I/O ports 0C0H to 0C7H of 8086. The 8259A PIC is set up to function at interrupt vectors 0E6H-0EDH and operates with rotate on nonspecific end of interrupt, specially fully nested mode and will trigger on level triggered mode. The 8259A PIC is decoded at I/O ports 72H and 73H. [See the necessary registers & command words at the end; Fig: 4(a-f) & Fig: 5(a-f)] **20**

Question – 6

a. Describe five interrupt vectors that are identical in all Intel microprocessor family members. **5**

b. Shortly describe the cast ineffective memory interfacing technique that improves the speed of a system in 80386. **7**

c. How is instruction/data processed in 80286 microprocessors? **8**

d. Develop a procedure for arithmetic coprocessor 80x87 that will calculate the area of a cylinder (whole surface) and the result must be saved in a variable, CYL_AREA. Also show the status of the stack of 80x87 after each instruction is executed. **15**

[Area of a cylinder = $2\,\pi \times radius \times (radius + height)$.
Assume that, radius = 3.5m and height = 9m]

Question – 7

a. Define the program status word and data pointer register of 8051 microcontroller. **5**

b. What are the two ways of testing status register in 80x87 arithmetic coprocessor? Describe the 6 errors that are found in 80x87 compressor. **8**

c. What is framing error and overrun error in 16550? 8086 microprocessor receives 7 bit data to other end through 16550. While receiving 7 bit data, some of the bits may have changed due to noise or distortion. Now you need to write a procedure to check if there is any framing error or overrun error. If there is an overrun error while receiving data, BL register will contain "?" and if there is a framing error then BL register will contain "!"; otherwise, it will transfer received data to AL register. [See the necessary register at the end; Fig: 5 (a) & (d)] **10**

d. Design a cascade 8259A PIC, where three 8259A PICs send their interrupt signal to master 8259A PIC. Suppose, the master 8259A PIC is decoded at I/O ports 0501H and 0505H, and slave 8259A PIC is decoded at I/O ports 050BH and 050DH. Master is used for interrupt vectors 0B5H-0BCH and slave is used for interrupt vectors 30H-37H. The three slaves are connected to interrupt vector 0B6H, 0B9H and 0BCH, respectively as cascade input. [Only draw the connection diagram] **12**

Question – 8

a. Shortly describe the way of measuring time interval in 80186. **5**

b. Describe the addressing modes of 8051 microcontroller. **8**

c. What do you understand by poll status register and end of interrupt register in 80186? Draw the block diagram of programmable interrupt controller of 80186/80188. **10**

d. Suppose in 8259A, programmable interrupt controller has 8 interrupt request pins (IR0-IR7). Among them, three interrupt request pins are unmasked (IR3, IR6, IR7). STI instruction of 8086 is enabled. An invalid opcode interrupt occurs at IR3 pin and 8259A sends the interrupt to 8086. While 8086 is executing invalid opcode interrupt procedure, another interrupt occurs at IR7. How will 8259A handle this new interrupt and what will happen if another interrupt occurs at IR6? Also, draw the diagram of the operating scenario. **12**

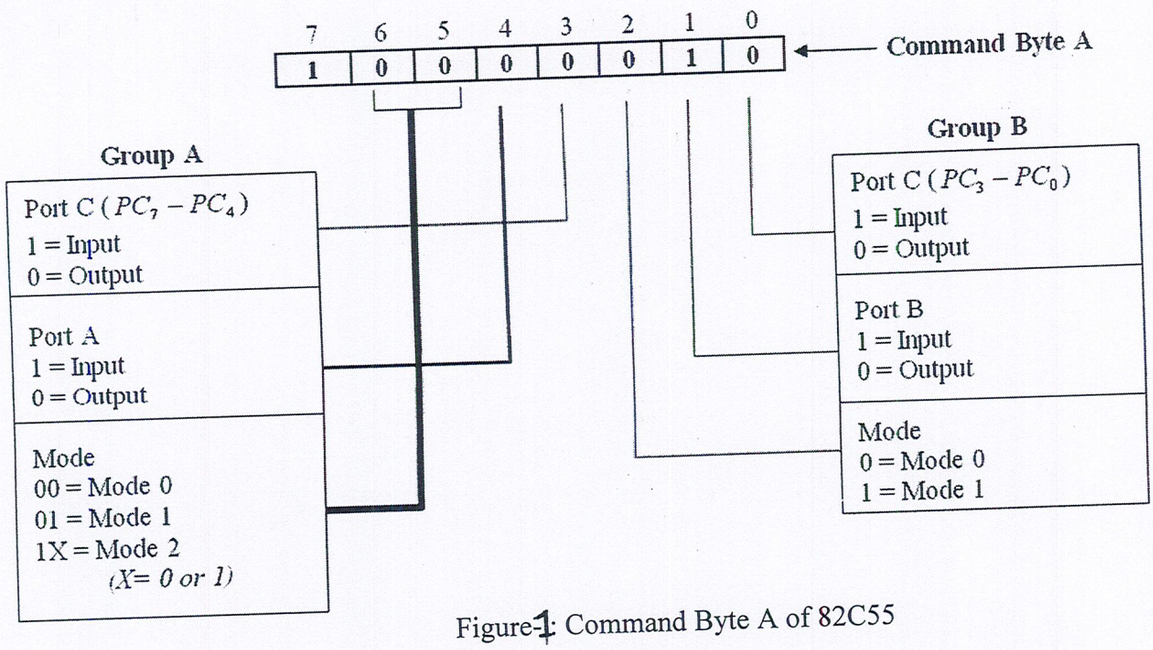


Figure-1: Command Byte A of 82C55

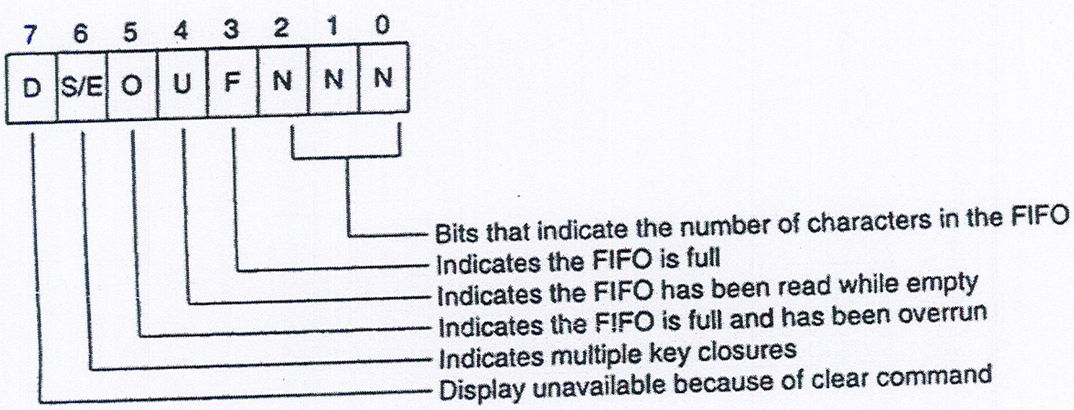


Figure-2 : FIFO Status Register of 8279

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

SC—Select Counter

SC1	SC0	
0	0	Select Counter 0
0	1	Select Counter 1
1	0	Select Counter 2
1	1	Read-Back Command (see Read Operations)

M—Mode

M2	M1	M0	
0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

RW—Read/Write

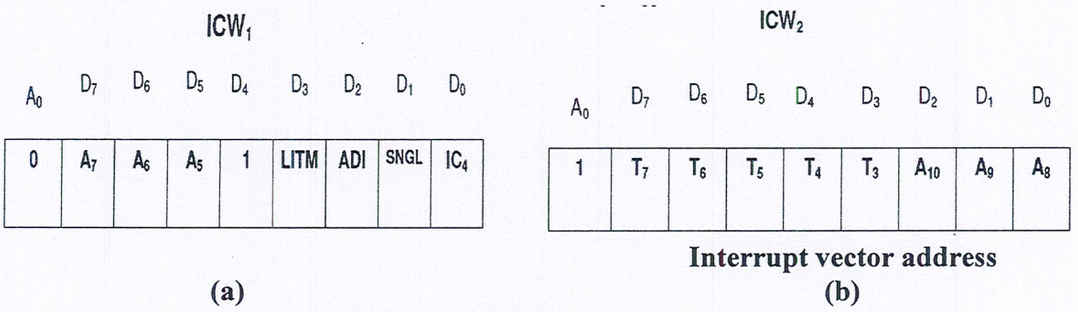
RW1	RW0	
0	0	Counter Latch Command (see Read Operations)
0	1	Read/Write least significant byte only
1	0	Read/Write most significant byte only
1	1	Read/Write least significant byte first, then most significant byte

BCD

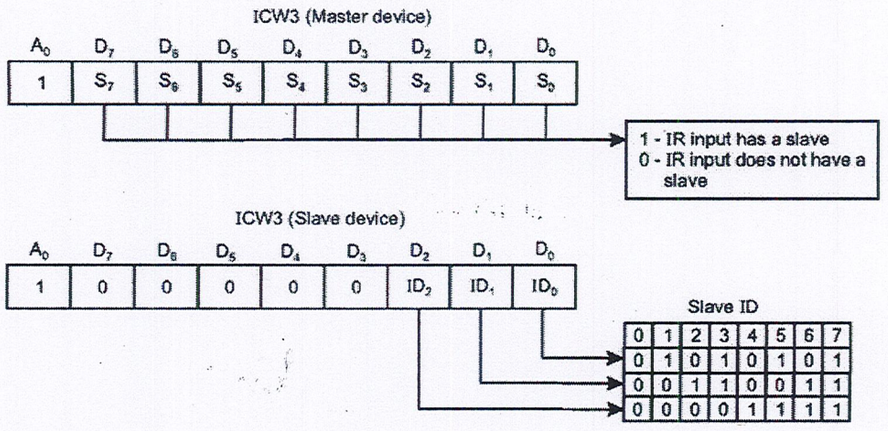
0	Binary Counter 16-bits
1	Binary Coded Decimal (BCD) Counter (4 Decades)

Figure-3 : Control Word Register of 8254

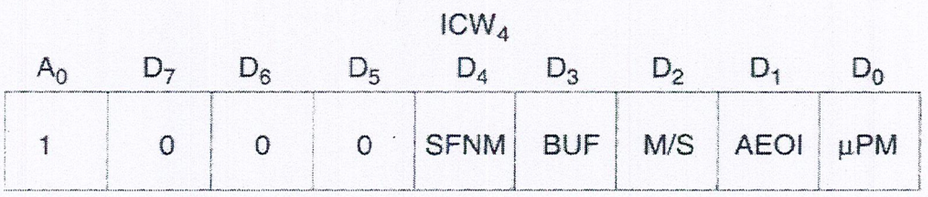
Nawrin



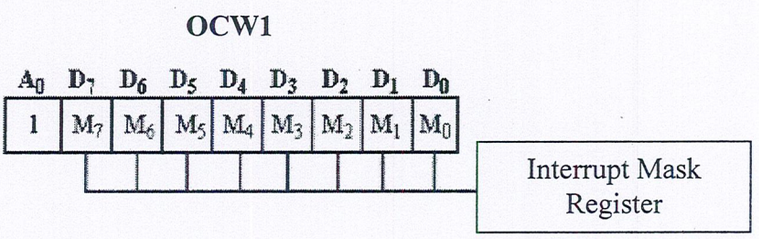
(a) (b)



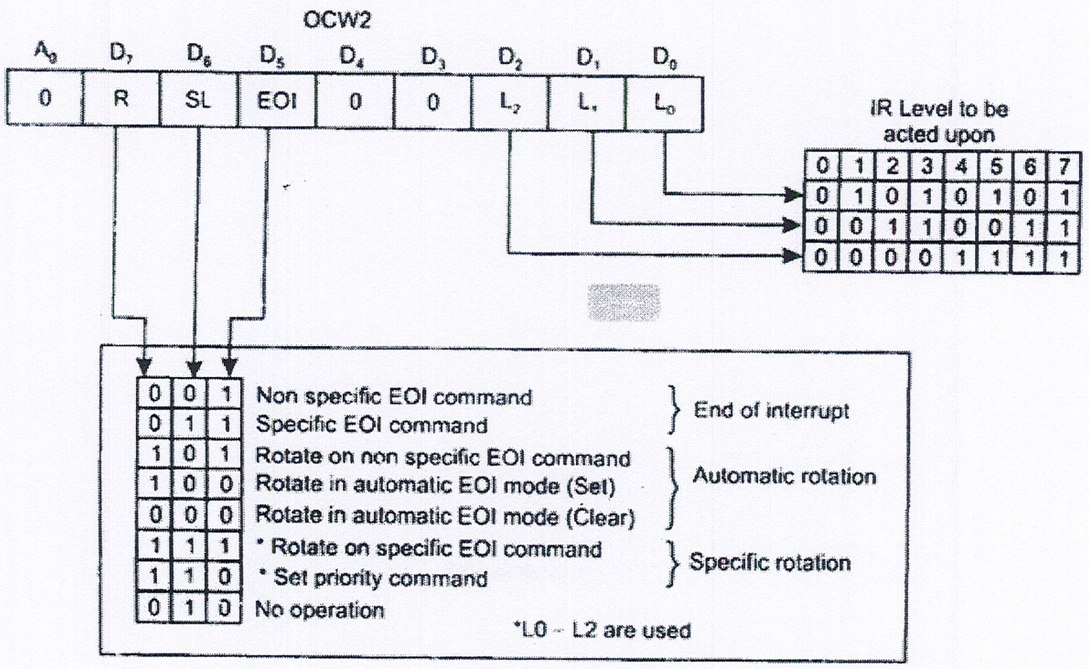
(c)



(d)



(e)



(f)

Fig:4 - Command Words of 8259A PIC

Anika

Register Selection in 16550

A_2	A_1	A_0	Function
0	0	0	Receiver buffer (read) and transmitter holding (write)
0	0	1	Interrupt enable
0	1	0	Interrupt identification (read) and FIFO control (write)
0	1	1	Line control
1	0	0	Modem control
1	0	1	Line status
1	1	0	Modem status
1	1	1	Scratch

(a)

Line Control Register

7	6	5	4	3	2	1	0
DL	SB	ST	P	PE	S	L1	L0

(b)

FIFO Control Register

7	6	5	4	3	2	1	0
RT1	RT0	0	0	DMA	XMIT RST	REVC RST	EN

(c)

Line Status Register

7	6	5	4	3	2	1	0
ER	TE	TH	BI	FE	PE	OE	DR

(d)

Interrupt Control Register

7	6	5	4	3	2	1	0
0	0	0	0	EM	EL	ET	ER

(e)

Interrupt Control Bits

Bit3	Bit2	Bit1	Bit0	Priority	Type	Reset Control
0	0	0	1	—	No interrupt	—
0	1	1	0	1	Receiver error (parity, framing, overrun, or break)	Reset by reading the line register
0	1	0	0	2	Receiver data available	Reset by reading the data
1	1	0	0	2	Character time-out, nothing has been removed from the receiver FIFO for at least four character times	Reset by reading the data
0	0	1	0	3	Transmitter empty	Reset by writing to the transmitter
0	0	0	0	4	Modem status	Reset by reading the modem status

(f)

Fig:5 – Registers of 16550 UART

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BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering
Student Group : 72 < Earned Credit Hour <= 108, Spring Final Examination: June 2019

Subject: CSE-305, Microprocessors and Microcontrollers

Time: 3.00 hours **Full Marks: 180**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. **Question-2** in **Section A** and **Question-5** in **Section B** are compulsory.
c. Answer any **Other TWO** questions from **EACH section**.
d. Figures in the margin indicate full **marks**.
e. Assume reasonable data if necessary.
f. **Symbols** used have their usual meanings.

SECTION-A

Question – 1

a. Distinguish between minimum mode and maximum mode operation of 8086. **10**

b. Distinguish between data addressing in memory by 80286 in real mode and protected mode. **20**

Question – 2

a. 80286 provides 'multitasking' and 'multiprogramming' environment. Discuss how 80286 protects tasks and program from each other? **12**

b. Discuss an 80386 segment descriptor. **18**

Question – 3

a. Discuss why 'paging' was introduced in 80386? What are its disadvantages? How was that solved? **15**

b. Discuss how a 32–bit logical address is converted into physical address in 80386? **15**

Question – 4

a. Discuss BCD and floating point data format of 80x87. **12**

b. Modify decimal 5.5 data to short [single precision] data format of 80x87. **5**

c. . Explain the bits of 80x87 status register. Exclude the condition code bits from the explanation. **13**

SECTION-B

Question – 5

a. Determine the mode set control word needed to initialize an 8255 as follows: **15**
Port A – bidirectional I/O
Port B – simple I/O
Port C – PC_0 and PC_1 as input.

b. List the different modes in which the ports of 8255 can be initialized. Discuss any two such modes. **15**

Question – 6

a. List and discuss the steps an 8086 will take when it responds to an interrupt. **12**

b. List and elaborate the different dedicated inputs used by 8086. **18**

Question – 7

a. 55H was sent to the control register of an 8254. Determine the operation of 8254. **15**

b. Explain the bits of 16550 line control register. **15**

Question – 8

a. 8259 can handle 8 interrupting devices. Design a system using 8259s that can handle 50 interrupting devices. **10**

b. 11H was sent as ICW1 to an 8259. Determine whether ICW3 and ICW4 are needed to complete the initialization of 8259. **5**

c. 00H was sent to an 8279 as control word. Explain the operation of the 8279. **15**

BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering
Student Group : 72 < Earned Credit Hour <= 108, Final Examination (Spring) :Feb 2021

Subject:CSE-305, Microprocessor, Microcontroller and assembly language

Total: 2.00 hours **Full Marks: 240**
Section A : 1.00 hour **Section A : 120**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. **Question – 1 and Question – 4 (Viva Voce)** in **Section A**are compulsory.
c. Answer any **OTHER ONE** question from this section **(From Q - 2 & Q - 3)**.
d. Figures in the margin indicate full **marks**.
e. Assume reasonable data if necessary.
f. **Symbols** used have their usual meanings.

SECTION-A

Question – 1 (Compulsory)

a. Determine the value of register AX after executing each of the following instructions sequentially **15**

```
main proc
  MOV AX,XXXXh ; where XXXX is the last four digits of your
                 student ID
  ADD AX,AX
  SUB AX,YYYYh ; where YYYY is the first four digits of your
                 student ID
  NEG AX
  INC AX
  AND AX,8000h
  NOT AX
main endp
```

b. Take a decimal number n as input and find out whether it is a palindrome or not. [For example, if the input is 15351 the program should produce the output message showing that "The given number is palindrome"] **15**

c. Define the segment registers of 8086 microprocessors. Determine the physical address of a memory location given by YYYYh:XXXXh where YYYY is the first four digits of your student ID and XXXX is the last four digits of your student ID. **10**

Question – 2

a. Suppose AL contains the binary value of 1XX where XX is the last two digits of your student ID and CF=1. Give the new contents of AL and CF after each of the following instructions is executed. Assume the preceding initial conditions for each part of this question. **15**

i. SHL AL,1
ii. SHR AL,1
iii. ROR AL,CL if CL contains 3
iv. SAR AL,CL if CL contains 2
v. RCL Al,1

b. Write a procedure REVERSE in assembly language that will reverse an array of N elements. The procedure is entered with SI pointing to the array and BX has the number of words N. **15**

c. How to connect a LED on Arduino? Write down the sketch for blinking an LED. **10**

Question – 3

a. If it is a legal instruction, give the values of DX & AX after each of the following instructions or tell if overflow occurs. **15**

i) MUL BX, if AX contains 000Xh and BX contains 0003h where X is the last digit of your student ID.

ii) MUL CX, if AX contains 00FFh and CX contains YYYYh where YYYY is the first four digit of your student ID

iii) IMUL CX, if AX contains 000Xh and CX contains FFFFh where X is the last digit of your student ID.

iv) DIV BX, if DX contains 0000h, AX contains XXXXh and BX contains YYYYh where XXXXh is the last four digit of your student ID and YYYYh is the first four digit of your student ID

v) IDIV BX, if DX contains FFFFh, AX contains FFFCh and BX contains 0003h

b. Suppose the following declaration have been made:
String1 db CSE-18
String2 db MIST-

Write an assembly code to move String1 to the end of String2, producing the string MIST-CSE-18. 15

c. Give a logic instruction to do each of the following: 10

i. Clear the even-numbered bits of AX, leaving the other bits unchanged.

ii. Set the LSB and MSB of BL, leaving the other bits unchanged

iii. Complement the MSB of DX, leaving the other bits unchanged

iv. One's complement of AX

Question – 4 (Compulsory) **40**
Marks allotted for Viva 40

BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering
Student Group :72 < Earned Credit Hour <= 108, Final Examination (Spring) :Feb 2021

Subject:CSE-305. Microprocessor, Microcontroller and Assembly Language

Total: 2.00 hours **Full Marks: 240**
Section B : 1.00 hour **Section B : 120**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. **Question – 5 and Question – 8 (Viva Voce)** in **Section B** are compulsory.
c. Answer any **OTHER ONE** question from this section **(From Q - 6 & Q - 7)**.
d. Figures in the margin indicate full **marks**.
e. Assume reasonable data if necessary.
f. **Symbols** used have their usual meanings.

SECTION-B

Question – 5 (Compulsory) **4x10=40**

a. Explain, with uses, the following 8086 signals:
   (i) ALE
   (ii) $\overline{BHE}$
   (iii) MN/$\overline{MX}$
   (iv) RESET

Question – 6

a. Distinguish between data addressing in memory by 80x86 in real and protected mode. **25**

b. Illustrate virtual 8086 mode operation of 80386 microprocessor. **15**

Question – 7

a. What is BIST in 80386? **10**

b. List and discuss the steps an 8086 will take when it responds to an external interrupt. **12**

c. Illustrate how an 8086 finds the address of an interrupt service routine (ISR) after an interrupt occurs. **8**

d. List the dedicated interrupts used by 8086 and elaborate any two of them. **2x5**

Question – 8 (Compulsory) **40**
Marks allotted for Viva 40

BANGLADESH UNIVERSITY OF PROFESSIONALS

Military Institute of Science and Technology

B.Sc. in Computer Science and Engineering, Term Final (Spring)Examination 2021: July-Aug 2021

Student Group: 72 < Earned Credit Hours ≤ 108

Subject: CSE-305, Microprocessor, Micro-controllers and Assembly Language

Total : 3.00 hours (For Physical Examination)

Full Marks : 240
Section A : 120

INSTRUCTIONS - PHYSICAL EXAMINATION

a. **Use SEPARATE answer scripts for each section.**
b. **Question - 4** in **Section A** is compulsory.
c. Answer any **OTHER TWO** questions from **Section-A**.
d. Figures in the margin indicate full **marks.**
e. Assume reasonable data if necessary.
f. **Symbols** and **abbreviations** used have their usual meanings.

SECTION-A

Question – 1

a. Consider the two binary numbers: **20**
1 1 0 1 0 1 1 0
1 1 1 0 1 1 0 1
For addition and subtraction, find out the FLAGS status.

b. Write a program in Assembly which will take three inputs from keyboard. Show that resultant value will be over at least (≥20), after adding all 3 numbers. **20**
For example:
1st value: 7
2nd value: 8
3rd value: 5
Result: 20

Question – 2

a. Consider the following sequence of bits: **20**
1 1 0 1 0 1 1 0
i. Now perform ROL using CL=5, where CF=1.
ii. Perform RCR using CL=5, where CF=1.

b. Assume you have a physical address B1128h and offset 6A08h. What will be the segment value? **10**

c. Give an example to show both overflows occurred. **10**

Question – 3

a. Write a program in Assembly to display the following pattern. For example: n = 5 it will print the below: **20**

```
    1
  2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

You need to take 'n' as an input.

b. Consider the following program: **20**

```
MOV AX, 5
MOV BX, 6
MOV CX, 7
MOV DX, 8
PUSH AX
```

```
ADD AX, CX
PUSH CX
PUSH AX
POP DX
SUB DX, BX
DEC DX
PUSH DX
POP AX
POP BX
POP CX
```

Now, draw stacks (whenever required) and show contents of relevant registers after each of the instruction above.

Question – 4 (Compulsory)

a. Give the new value of DX and AX (in decimal/ hexa) and CF/ OF for each of the following instructions: **08**

i) IMUL CX; where AX = 0100h and CX = FFFFh
ii) IDIV BX; where AX = FFFDh, BX = 02h, DX = FFFFh
iii) DIV BX; where AX = FFFDh, BX = 2h, DX = FFFFh

b. Write a program in Assembly language to replace each lowercase letter in the following string by its uppercase equivalent. **12**
MSG DB 'this is a message'
Recall, ascii value of 'A' is 65 and 'z' is 122.

c. Suppose **20**
AX = 0500h, BX =1000h, SI = 1500h, DI =2000h
offset 1000h contains 0100h
offset 1500h contains 0150h
offset 2000h contains 0200h
offset 3000h contains 0400h
offset 4000h contains 0300h and
BETA is a word variable whose offset dress is 1000h.

Now for each of the following instructions, if legal, give the source offset address or register and result stored in the destination:

```
 i.   MOV DI, SI            vi.   ADD [SI], [DI]
ii.   MOV DI, [DI]          ii.   ADD BH, [BL]
ii.   ADD AX, [SI]          ii.   ADD AH, [SI]
iv.   SUB BX, [DI]          ix.   MOV AX, [BX+DI+BETA]
 v.   LEA BX, BETA[BX]       x.   SUB BETA [DI], AX
```

BANGLADESH UNIVERSITY OF PROFESSIONALS

Military Institute of Science and Technology

B.Sc. in Computer Science and Engineering, Term Final (Spring)Examination 2021: July-Aug 2021

Student Group: 72 < Earned Credit Hours ≤ 108

Subject: CSE-305, Microprocessor, Micro-controllers and Assembly Language

Total : 3.00 hours (For Physical Examination)

Full Marks : 240
Section B : 120

INSTRUCTIONS - PHYSICAL EXAMINATION

a. **Use SEPARATE answer scripts for each section.**
b. **Question - 8** in **Section B** is compulsory.
c. Answer any **OTHER TWO** questions from **Section-B**.
d. Figures in the margin indicate full **marks.**
e. Assume reasonable data if necessary.
f. **Symbols** and **abbreviations** used have their usual meanings.

SECTION-B

Question – 5 **20**

a. Write down a program in both Assembly language and corresponding Machine Language to determine the result in SPA1 for the following arithmetic expression:

$$-18 + (12 - 10) - (6 - 28)$$

You need to specify both the address and contents (in RAM) for both the languages. **20**

b. Show the timing diagram (T1 – T6) for SAP1 SUB routine, specifying all the control signals:

$$\text{CON} = C_P E_P \overline{L_M} C E \overline{L_I} \overline{E_I} \overline{L_A}\ E_A S_U E_U \overline{L_B}\ \overline{L_O}$$

Question – 6

a. Assume, the following declaration have been made so far: **20**

```
STR2 DB 'CDEF'
STR1 DB 'ABXY'
     DB 4 DUP(?);Now, STR1 can have at most 8 characters
```

Now write instructions to move STR2 in the middle of STR1, producing the string (STR1) "ABCDEFXY". You may use temporary strings as necessary. You must use string instructions to write this program.

b. Write down the output of the following program: **12**

```
.CODE
main proc

mov ch,1
All_:
MOV AL,ch

cmp al,2
jl FALSE
je TRUE

MOV BL,02H
MOV DX,0000H
MOV AH,00H

L1:
DIV BL
CMP AH,00H
JNE NEXT
jmp FALSE

NEXT:
INC BL
;reset for division
MOV AX,0000H
MOV DX,0000H
MOV AL,ch
CMP BL,ch
JL L1

TRUE:
MOV AH,02H
mov dl,
int 21h
mov dl,ch
add dl,48
INT 21H
mov ah,0
JMP EXIT

FALSE:

EXIT:
inc ch
cmp ch, 10
jl All_

MOV AH,4CH
INT 21H
main endp
end main
```

c. Use the addressing modes used in Array addressing to write appropriate instructions equivalent to each of the following string instructions: **08**

i. CMPSB
ii. SCASB

Note that the flags should reflect the result of the comparison. You may use AL for temporary storage, and assume DF = 0. Also, assume that S1 and DI already have the offset addresses of the source string and destination string respectively.
[Hint: for MOV SB:

```
MOV AL, [SI]
MOV [DI], AL
INC SI
INC DI          ]
```

Question – 7

a. Explain with necessary figure how address and data bus of 8086 are demultiplexed using its interface circuitry. **20**

b. What is the physical address space (range of address) in IVT of 8086? Calculate the memory address/ location for the interrupt INT 5H in IVT. You must specify the addresses for CS and IP. **12**

c. Why exactly 4 bytes are needed for each interrupt in IVT? Write down he main difference(s) between IVT and ISR. **08**

Question – 8 (Compulsory)

a. Show the BUS timing diagram for an 8086 I/O Read operation, specifying the signals for the following pins: **20**

i) $A_{19} - A_{16}/ S_6 - S_3$
ii) $AD_{15} - AD_0$
iii) $\overline{RD}$
iv) $\overline{WR}$
v) $DT/\overline{R}$
vi) ALE
vii) $MN/\overline{MX}$
viii) $M/\overline{IO}$

b. Write down the basic functions of the following pins of 8086: **20**

i) $MN/\overline{MX}$
ii) NMI
iii) $READY$
iv) $\overline{TEST}$

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Student Group: 72 < Earned Credit Hours ≤ 108

Subject: CSE 305, Microprocessors, Micro-controllers and Assembly Language

Time: 4.00 hours **Full Marks: 240**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. **Question-1** in **Section-A** and **Question-5** in **Section-B** are compulsory
c. Answer any other **THREE** questions out of **FOUR** from each section.
d. Figures in the margin indicate full **marks**.
e. Assume reasonable data if necessary.
f. **Symbols** and **abbreviations** used have their usual meanings.

SECTION-A

Question 1 **(Compulsory)** **Marks**

a. Analyze the protection scheme of 80286 among multiple tasks or users. **14**

b. Suppose an 80286 segment descriptor is described as $(0000912020141234)_{16}$. Analyze
   (i) The location of the segment in memory,
   (ii) Size of the segment,
   (iii) Who can access this segment and
   (iv) Whether this is a code or data or stack segment. **7+7+7+5 =26**

Question 2

a. Once you are in "protected mode" in 80286, you cannot come back to "real mode" without "rebooting" the processor. How 80386 handles this problem? **16**

b. Analyze a segment descriptor of 80386. **24**

Question 3

a. List the specialties of port C in comparison to port A or Port B of an 8255. **12**

b. The control register of an 8255 contains $(FF)_{16}$. Analyze the operation of the 8255 ports. **16**

c. Compare the "minimum mode" and "maximum mode" of operation of an 8086. **12**

Question 4

a. List and describe the steps INTEL processors will take when they respond to an external interrupt. **16**

b. Analyze why Pentium is known as **"**Super scalar**"** processor. **8**

c. Illustrate the differences of page memory management of 80386 and Pentium. **16**

SECTION-B

Question 5 **(Compulsory)**

Scenario :

Entrepreneur Mr. X runs a modest business. His products are safe at normal temperatures (below 25 degree Celsius), but their quality deteriorates at temperature above 30 degrees Celsius. Mr. X cannot afford to have the air conditioning on all the time because of his poor financial situation and rising utility costs. Therefore, he seeks a low cost device that can alert him when to active the air conditioning.

Now, he wants to design and developed a microcontroller-based (Arduino) System that will do the following :

(i) Above 30 degree Celsius the Red LED will blink.
(ii) Between 25 degree and 30 degree a Yellow LED will turn ON (in low intensity).
(iii) Under 25 degree Celsius a Green LED will turn ON.

You need to answer the followings based on the above scenario :

a. Determine the hardware requirements. **7**

b. Draw the hardware connections. **13**

c. Write down the code (Arduino Code) accordingly to the hardware connection of b. **20**

Question 6

a. Illustrate with examples how overflow is detected for both signed and unsigned numbers in 8086. **12**

b. Construct an assembly program that will prompts the user to enter a character, and on subsequent lines prints its ASCII code in binary, and the number of 1 bit in its ASCII code. **13**

c. Construct a program in assembly language to display the following pattern. For example, n = 5 will print the below where n is the user input. **15**

```
1 2 3 4 5
2 3 4 5
3 4 5
4 5
5
```

Note that, the range of n is 0 to 5.

Question 7

a. Describe the effect of MUL and IMUL on the status flags of 8086. For the following instructions find out the new value of AH, AL and CF/OF. **16**

(i) MUL BL; if AL contains FFh and BL contains 80h
(ii) IMUL BL ; if AL contains FFh and BL contains 80h

b. Write a procedure FIBONACCI that will compute the nth term of the Fibonacci sequence for a positive integer n. The procedure should receive n in CX and return the nth term of the fibonacci sequence in AX. Assume that there will be no overflow. **16**

c. Briefly discuss the following instructions with appropriate examples: **8**

(i) PUSHF
(ii) POPF

Question 8

a. Illustrate with examples how a number can be checked whether it's odd or even without division or addition or subtraction instructions in 8086 using assembly language. **12**

b. Discuss how an instruction is executed in 8086 in terms of the execution steps. **12**

c. Using only MOV, ADD, SUB, INC, DEC and NEG change the following high level language assignment statements into low level assembly language. Assume, A, B, C are Word variables. **16**

(i) A = - A + B – 1
(ii) B = 3B + 2A -7
(iii) C = - (C + 1)
(iv) B = A – B
(v) C = 2(A+B) – 4C

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Subject: CSE 305, Microprocessor, Micro-controller and Assembly Language

Time: 3.00 hours **Full Marks: 180**

INSTRUCTIONS:

a. Use **SEPARATE** answer scripts for each section.
b. **Question-1** in **Section-A** and **Question-5** in **Section-B** are compulsory.
c. Answer any other **TWO** out of the remaining **THREE** questions from each section.
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e. Assume reasonable data if necessary.
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SECTION-A

Question 1 (Compulsory)

a. Briefly describe the interrupts of 8086 predefined by INTEL. **12**

b. List and describe the steps INTEL microprocessor take when an external interrupt arrives. **12**

c. What address in the interrupt-vector table are used for a Type-1 interrupt? **6**

Question 2

a. Analyze how protection among tasks or users are ensured in 80286. **10**

b. Suppose DS [Data Segment] register contains $(OOFF)_{16}$ which points to a segment descriptor which is described in memory as$(000091202114 5678)_{16}$. **(6+6+6+2=20)**
Analyze
(i) Segment start address,
(ii)Size of the segment in bytes,
(iii) Who can use this segment and
(iv) Type of the segment (code/data/stack).

Question 3

a. In 80286, returning from "protected mode" to "read mode" is not allowed and not possible without "rebooting". How 80386 made this possible? **12**

b. Analyze 80386 segment descriptor. **18**

Question 4

a. Explain why Pentium is known as 'superscalar' microprocessor. **6**

b. Illustrate how paging memory management of Pentium is different from that of 80386 processor. **12**

c. The control register of an 8255 contains$(F7)_{16}$. Analyze the operations of 8255 ports. **12**

SECTION-B

Question 5 **(Compulsory)**

Mr. X owns a small green house where he grows temperature sensitive plants. To automate the environment control, he want to build a system using two microcontroller boards. One microcontroller, placed in his control room, will act as the Host, while the other will be inside the greenhouse as the local device. The host microcontroller will be connected to a LED and will send command to the Local microcontroller through UART communication. The local microcontroller, equipped with an LM35 temperature sensor and a servo motor, will read the temperature of the greenhouse and send it back to control panel. If the temperature exceeds a set threshold, the host will instruct the local to adjust the vents via the servo motor, maintaining an ideal environment for the plants.
Now, Mr. X wants to design and develop a microcontroller based system that will do the following:
(i) If the temperature of the greenhouse goes under 25 degree, then LED will turn off and the vents will be adjusted to 90º.
(ii) If the temperature of the greenhouse goes above 25 degree, then the LED will turn on and the vents will be adjusted to 180 º.

Note that, LED is connected with the Host and the vent is connected with the Local microcontroller.
You need to answer the followings based on the above scenario:

a. Identify the hardware requirements **5**

b. Draw the hardware connections. **10**

c. Write down the code accordingly to the hardware connection of 5(b). **15**

Question 6

a. Describe the effect of MUL and IMUL on the status flags of 8086. **15**
For the following instructions find the new value of AH, AL and CF/OF.
(i) MUL BL; if AL contains 80H and BL contains FFH.
(ii) IMUL BL; if AL contains 80H and BL contains FFH.

b. Describe the following instructions with appropriate examples: **15**
(i)PUSH
(ii)POP
(iii)CALL

Question 7

a. Find out the value of DL register and carry Flag status after the execution of each code snippet: **15**

(i)
```
MOV CL, 03H
MOV DL, 18H
SHL  DL, CL
```

(ii)
```
MOV CL, 03H
MOV DL, -18H
SAR DL, CL
NEG DL
```

b. Construct a program in assembly language to display the following patterns. For example, n = 5 will print the below pattern where n is the user input. Note that, the range of n is 0 to 5. **15**

1
2 3
3 4 5
4 5 6 7
5 6 7 8 9

Question 8

a. Write an assembly program that lets the user type some text, considering words separated by spaces, ending with a carriage return, and displays the text in the same word order as entered, but with the letters in each word reversed. **15**

Sample Input-Output:
Enter a string: this is a test
Output: siht si a tset

b. One of the fundamental differences between an Arduino and Raspberry Pi is the presence of an operating system. Describe the implications of using an OS (like Linux) on Raspberry Pi compared to the Base- metal programming Approach on Arduino. **10**

c. Describe how an Arduino Uno sends an analog output. **5**

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SECTION-A

Question 1 **(Compulsory)**

a. What are the predefined interrupts in 8086? Briefly describe them. **16**

b. List and describe the steps INTEL microprocessor take when an interrupts arrive. **16**

c. Calculate the address from where CS and IP can be obtained for interrupt type 2. **8**

Question 2

a. The control register of an 8255 contains $(F7)_{16}$. Analyze the operations of 8255 ports. **16**

b. Explain why Pentium is known as 'superscalar' microprocessor. **8**

c. Illustrate the difference of paging memory management of Pentium to that of 80386. **16**

Question 3

a. Analyze how protection among tasks or users are ensured in 80286. **14**

b. Suppose a segment register contains $(OOFF)_{16}$ which points to a segment descriptor which is described in memory as$(0000912021145678)_{16}$. **(8+8+7+3)**
Analyze
(i) Segment start address,
(ii)Size of the segment in bytes,
(iii) Who can use this segment and
(iv) Type of the segment (code/data/stack).

Question 4

a. Discuss the 'virtual 8086' mode of operation of 80386. **16**

b. Analyze 80386 segment descriptor. **24**

SECTION-B

Question 5 **(Compulsory)**

Mr. X owns a small green house where he grows temperature sensitive plants. To automate the environment control, he want to build a system using two microcontroller boards. One microcontroller, placed in his control room, will act as the Host, while the other will be inside the greenhouse as the local device. The host microcontroller will be connected to a LED and will send command to the Local microcontroller through UART communication. The local microcontroller, equipped with an LM35 temperature sensor and a servo motor, will read the temperature of the greenhouse and send it back to control panel. If the temperature exceeds a set threshold, the host will instruct the local to adjust the vents via the servo motor, maintaining an ideal environment for the plants.

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Note that, LED is connected with the Host and the vent is connected with the local microcontroller.

You need to answer the followings based on the above scenario:

a. Identify the hardware requirements **7**

b. Draw the hardware connections. **13**

c. Write down the code accordingly to the hardware connection of 5(b). **20**

Question 6

a. Describe the effect of MUL and IMUL on the status flags of 8086. **20**
For the following instructions find the new value of AH, AL and CF/OF.
(i) MUL BL; if AL contains 80H and BL contains FFH.
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b. Describe the following instructions with appropriate examples: **20**
(i)PUSH
(ii)POP
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```
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```

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```
1
2 3
3 4 5
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```

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SECTION-A

Question 1 (Compulsory)

a. Elaborate the processes and events that will take place when an INTEL microprocessor received a high signal on its INTR pin. **12**

b. Briefly discuss the pre-defined interrupts of 8086. **12**

c. Calculate the address of Type 4 interrupt of 8086. **6**

Question 2

a. List the differences in characteristics between Port A and Port C of an 8255. **10**

b. Compute the mode set control word for the following configuration of an 8255:
Port A in bidirectional and Port B in hand-shake input. **10**

c. Suppose a segment register of 80386 points to a segment descriptor as $(20\ C0\ F0\ A1\ 23\ 45\ 67\ 89)_{16}$. Find (i) The starting address and (ii) Size of the segment **10**

Question 3

a. Explain why Pentium is known as ‘super-scalar’ processor. **7**

b. Explain why 8086 is known as 16-bit processor whereas Pentium is known as 32-bit processor. **7**

c. Differentiate between Pentium and 80386 processor’s paging scheme. **16**

Question 4

a. Explain the ‘minimum’ and ‘maximum’ mode of operations of 8086. **6**

b. Differentiate the usage of segment registers of 80286 in real mode and protected mode. **12**

c. Elaborate how 80286 ensures protection between users and tasks. **12**

SECTION-B

Question 5 (Compulsory)

Azmain Taki, the owner of a manufacturing company, plans to introduce a mini robotic crane prototype for use in an automated assembly line. He has assigned you, as an embedded systems engineer, to design the system using an ATmega328P based microcontroller.

The crane system must meet the following requirements:

- A base servo motor rotates the crane arm within 0° - 180°.
- A claw servo motor can open (0°) and close (90°) to grip objects.
- A joystick module is provided for operator input: moving the joystick left/right rotates the base.
- Pressing the joystick button toggles the claw open/close.
- Two indicator LEDs are placed on either side of the crane to indicate the current rotation direction (left LED ON when moving left, right LED ON when moving right).
- The system is powered by a regulated 5V supply, with all grounds common.

Azmain Taki expects a working prototype with both the hardware prototype interfacing and the software program implemented correctly. Additionally he expects you to maintain documentation of the used microcontroller.

TASKS:

a. Sketch a complete interfacing diagram of the system, showing connections between the microcontroller, joystick module, servos, LEDs, and power supply. Clearly label the ports and pins used. **10**

b. Write a program that: **10**

- Reads joystick input (X-axis via analog input & button as digital input).
- Rotates the base servo proportionally from 0°-180° based on joystick position.
- Lights up the corresponding LED according to base movement direction.
- Toggles the claw servo between open (0°) and closed (90°) on each button press.

c. Sketch the Port Structured Diagram of ATmega328P microcontroller. **10**

Question 6

a. Below is the basic program structure of an assembly program. Explain each part of the program structure briefly. **10**

```
.MODEL SMALL
.STACK 100H
.DATA

.CODE
MAIN PROC

   MAIN ENDP
END MAIN
```

b. Verify the following instructions if it is legal or illegal. If illegal, explain why? **10**

i) MUL AL, AL
ii) MOV AK, VAR1
iii) MOV AL, 10000000
iv) ADD AL, DX
v) NEG VAR1

c. Write assembly instructions to formulate the following equation for unsigned numbers. Use as much variables as you want. **10**

$$Y = (M^3 - 2N^2) + \left(\frac{P \times Q}{R^2}\right) - (\left(\frac{S \times T}{16}\right)\%5)$$

Question 7

a. Write an assembly program that takes *Income* as input (Decimal) (max value 100) and calculate *income_tax* based on the following rules: **20**

i) If income < 0, print "Invalid"

ii) Otherwise, compute tax as:

- for income 0 tk – 25 tk range: No tax
- for income 26 tk – 50 tk range: 25% of the amount above 25.
- for income 51 tk – 100 tk range:
  25% of the first 25 tk above the base 25 tk
  +
  50% of the amount above 50 tk
- for income above 100 tk:
  25% of the slab 25 tk – 50 tk
  +
  50% of the slab 51 tk – 100 tk
  +
  75% of the slab above 100 tk

iii) After computing the tax, apply a rebate after taking age as input:

- If age >= 60, reduce the tax by 25%.
- However, the maximum rebate allowed is 25 tk.

iv) Finally print the final payable tax in Decimal Number.

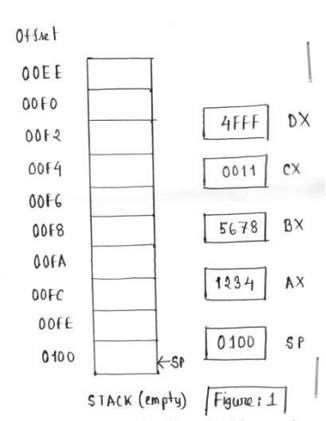
b. Write the algorithm for Binary Output and illustrate the full process showing the state of the register after each iteration. Imagine the register and input being 8 bits. **10**

Question 8

a. Write an assembly program that will take a decimal number as input and calculate the factorial of that number and print it. **10**

b. The initial state of registers and stack of a system is given in figure 1. Now after executing the following instructions, sketch the final state of the registers and the stack. **10**

```
PUSH AX
PUSH BX
POP CX
ADD AX, CX
PUSH DX
SUB AX, BX
POP BX
PUSH CX
PUSH AX
ADD AX, BX
POP DX
SUB BX, CX
PUSH BX
POP AX
POP CX
```



c. Describe the following instructions with appropriate examples: **10**

i) IMUL

ii) DUP Operator

iii) CALL

iv) RET